SuperAdmin Manual

FinOps SaaS · CSCloudSolutions

Version 2.0 · August 2026

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📘 SuperAdmin Manual — FinOps SaaS (CSCloudSolutions)

Version: 2.0 (detailed) **Language:** English **Last updated:** August 2026 **Audience:** CSCloudSolutions internal team only (SuperAdmin). Includes everything in the User Manual plus platform administration flows (tenant onboarding, customer onboarding, sales tracking).

Recent updates (August 2026)

- **Compute FinOps Cockpits & Serverless Efficiency:** Added resolution cockpits for **Virtual Machines** (`/intelligence/computo/avm`) with active compute vs. persistent storage cost separation (deallocated waste detection), AHUB licensing, B-series rightsizing, 8x5 schedules, and abandoned VM disposal; **Function Apps** (`/intelligence/computo/fapps`) with invocation telemetry, GB-seconds, dependency audits (Storage/App Insights), and sampling/downgrade playbooks; **App Services** (`/intelligence/computo/waas`) with app density and packing; and **VMSS** (`/intelligence/computo/vmss`) with autoscale and Spot pricing.
- **Responsive Popovers Standardization (`InfoTooltip`):** 100% responsive popovers with institutional `#1B2A41` background ensuring zero text clipping across any viewport.
- **New SuperAdmin panel:** `Partner Center Alerts` (`/superadmin/partner-alerts`) to monitor PAL/CPOR status by tenant and detect recent events.
- **New Azure Integration Services (iPaaS) module:** `Intelligence → Azure Integration Services` now includes tabs for Logic Apps, APIM, Service Bus, Event Grid, Event Hubs, and ADF, with a dedicated Enterprise Connectors section for Logic Apps.
- **PAL operational automation:** when partner association is approved, events are logged and automatic retry (cron) updates status and detail.
- **Enterprise AI (Azure IA):** global configuration now stores and tests **endpoint URL** + deployment in addition to API key, with backward-compatible fallback.
- **Cross-cutting FinOps/CMP table directive:** formalized for current and future tables (base filters, A-Z/Z-A/cost sorting, 15/30/45/60 pagination, responsive full-width layout, and resizable columns).
- **Security and Monitoring standardized:** both modules now match the same style used in Databases and Compute, including pagination and resource/region/type/resource-group filters.
- **New Security operational cron:** `prewarm-security-finops` added to Terraform cron maps (staging/prod) to prewarm `defender` and `security/service-cost` .
- **Key Vault Hardening & Network Isolation:** Production Key Vault protected with a dedicated Private Endpoint (`cscs-finops-prod-wus2-kv-pe`) in `snet-pe` , firewall set to `default_action = Deny` , and automated ephemeral runner opening in GitHub Actions.
- **Exhaustive Daily Prewarm (4:00 AM) & Async Poll Architecture:** Deployed `prewarm-daily` cron (`0 4 * * *` , 07:00 UTC) to pre-calculate and cache to Redis (26h TTL) full KQL audits, zombies, whiteboard, MTD/historical costs, inventory, tags, FinOps maturity, and scorecards. Migrated `prewarm-databases` and `prewarm-compute` Container App Jobs to `async_poll = true` protocol to eliminate 240s Ingress timeouts.
- **Azure SQL Elastic Pool Billing Correction:** Assigned `$0.00 (In Pool)` cost to Elastic Pool member databases, preventing duplicate multi-database compute multiplication and properly billing at the Elastic Pool resource level.
- **AI Cost Analytics hardening:** fixes for real-consumption display, daily MTD trend, and cache/data-source robustness for Microsoft Foundry.
- **Database metrics hardening:** Redis, MySQL, PostgreSQL, Cosmos DB, MongoDB, and SQL/Managed Instance now use per-metric fallback to prevent `N/A/unknown` dashboards under partial Azure telemetry.
- **Tabler-based FinOps visual refresh:** key modules in Intelligence, Consumption, Governance, Cleanup, Overview, and Copilot M365 are now standardized with no blue icon backgrounds.

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- **Staging pipeline hardening:** deployment now resolves subscription/RG/app/job dynamically, tracks migrations with the proper `job-execution-name`, and treats redirect health checks (`200/307/308`) as healthy.

Change-control directive (mandatory)

For agent-assisted tasks in this repository:

- **Do not push to `staging` or `main` without explicit user instruction.**
- **Do not merge into `main` without explicit user instruction.**
- Default mode is **local commits only** until an explicit promotion request is provided.

How to use this manual

Each section explains **what** the feature is, **who** can use it (role and subscription tier), and the **step-by-step usage flow** — concrete buttons, URL paths, form fields, and expected results. If you're a non-technical user, feel free to jump straight to the section you need: each one is self-contained.

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1. Getting Started

1.1. Access and login

The platform is a B2B SaaS. Authentication is integrated with **Microsoft Entra ID** (Azure Active Directory):

1. Go to the platform URL.
2. Click **"Sign in with Microsoft"**.
3. Authenticate with your corporate account. The platform automatically recognizes your Azure tenant and identity.
4. **Demo Mode:** if you want to try the platform without connecting your real environment, choose one of the preconfigured demo profiles from the main screen — they come with realistic simulated data and metrics, so you can explore every module risk-free. There are four demo tenants (one per tier).

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1.2. The onboarding wizard (first time)

If you're an administrator and this is your organization's first time using the platform, upon login the **Setup Wizard** (`/onboarding`) automatically opens — a 5-step linear wizard with a progress bar:

Step	What you do	Result
1. Welcome & Company Info	Confirm company name, primary cloud provider (Azure), display currency (USD/EUR/GBP), and timezone	Your initial preferences are saved
2. Connect Azure Subscription	Paste **Client ID**, **Client Secret**, and **Azure Tenant ID** from the Service Principal (generated with the PowerShell script provided by CSCloudSolutions) and click **"Validate"**	The system checks live whether the Service Principal has the minimum required roles; if any is missing, it shows exactly which one in red
3. Run First Data Sync	Click **"Run Sync"**	Fetches your first set of cost data from Azure (can take up to 60 seconds)
4. Create Your First Budget	Fill in name, monthly limit ($), and alert threshold (%)	Your first active budget is created
5. Set Up Notifications	Click **"Configure"** (opens `/admin/notifications` in a new tab)	Add at least one channel (email, Slack, or Teams) to receive alerts

You can **skip** any step and return later — the wizard keeps appearing until you complete or skip all 5. Progress (%) is calculated as `(completed + skipped) / 5 × 100` . If you prefer the advanced setup form instead of the wizard, `/admin/onboarding` remains available in parallel.

1.3. Onboarding a new customer (SuperAdmin flow)

If you're a CSCloudSolutions SuperAdmin onboarding a new tenant:

1. **Register the Tenant:** go to `/admin/tenants` → **Tenant Management** → enter the customer's Entra ID, business name, and initial Tier.
2. **Generate credentials:** once created in the database, go to `/admin/onboarding` → **Customer Onboarding**. The **Client ID** and **Client Secret** fields now appear, to paste the Service Principal credentials generated by the PowerShell script you shared with the customer.
3. **Tag commercial origin and commission (optional):** in each tenant's expanded panel in the Environments Directory, set **"Commercial Origin / Referred by"** and **"Commission (%)"** for internal commission settlement — visible only to SuperAdmins.
4. **Partner association (PAL / CPOR):** after credentials are loaded, the approve/decline block remains visible until the state is **linked (LINKED)**. If it ended in `FAILED` or `DECLINED` , you can retry without manual state reset.
5. **Secure Customer Session Impersonation:** from the Tenants table (`/admin/tenants`), each row features an **"Impersonate"** action button (`IconUserCheck`). Clicking it:
   - Sets an encrypted HTTP-only cookie `saas_impersonation_session` with a 4-hour lifespan.
   - Grants the SuperAdmin immediate visibility into the exact tenant workspace, subscriptions, and RBAC views.
   - Displays a fixed top banner (`z-[90]`) with delegation details and an instant **"Exit Impersonation"** return button.
   - Automatically writes an immutable audit record in `AuditTrailLogs` with `isImpersonated: true` and `executedBySuperAdmin` .

Azure roles the script assigns, by contracted tier:

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Tier	Built-in roles	Custom role
Professional (platform floor)	Reader, Cost Management Reader, Monitoring Reader, Billing Reader	—
Business	Professional + Tag Contributor	VM Start/Stop/Restart/Deallocate + tags
Enterprise	Business + Tag Contributor	Business + delete disk/snapshot/NIC/Public IP/NSG

> ⚠️ **The 4 base Professional roles are the absolute minimum** for the Real Consumption page to show data. If `Cost Management Reader` or `Billing Reader` is missing, Azure silently returns 0 rows.

> ℹ️ **AI Cost Analytics (Microsoft Foundry / Azure OpenAI)** uses the same base roles (`Reader` , `Cost Management Reader` , `Monitoring Reader` , `Billing Reader`): **no extra role is required**.

> ⚠️ **EA/MCA subscriptions** (Enterprise Agreement / Microsoft Customer Agreement) additionally require the customer's Billing Admin to assign `Enrollment Reader` or `Billing Account Reader` to the Service Principal at the billing account scope — the script can't do this automatically, it must be coordinated with the customer.

Verifying permissions were correctly assigned, after the customer runs the script:

- **Quick option (recommended):** `GET /api/admin/check-sp-roles?tenantId=<tenant-id>` — returns how many subscriptions have all required roles ✅, which are partial ⚠️, and which have none ❌, with the exact detail of what's missing in each.
- **Manual option:** in Azure Portal → Subscription → **Access control (IAM)** → **Role assignments** → filter by the App Registration `CSCloudSolutions-FinOps-Agent` and confirm the tier's roles appear.

Troubleshooting — "Real Consumption shows no data":

Error message	Cause	Fix
`NO_COST_PERMISSION`	Missing `Cost Management Reader`	Assign the role to the SP on that subscription
`NO_SUBSCRIPTION_ACCESS`	Missing `Reader`	Assign `Reader` or re-run the script
`SUBSCRIPTION_INACTIVE`	The subscription has no spend in the current month or the last 30 days	Verify it's the correct subscription
`NO_SUBSCRIPTIONS`	The SP sees no subscriptions	Assign `Reader` on at least one
`NO_CONSUMPTION`	Everything OK but no spend in the current cycle	Wait for the billing cycle to close or check another subscription

Troubleshooting — "AI Cost Analytics (Microsoft Foundry) shows 0 tokens / $0":

Even when models (e.g. gpt-5.1, gpt-5.3-codex) have real usage in Microsoft Foundry, the panel may stay at zero for several reasons. Use the **diagnostics endpoint** (no database or cron-log access required):

- `GET /api/intelligence/ai-analytics/diagnostics?tenantId=<tenant-id>` — returns, step by step: current rows in `AICostSnapshots` , visible subscriptions (with tier truncation), discovered `Microsoft.CognitiveServices/accounts` and their `kind` , the **metrics actually available** per account, a **live probe** of each token metric (series count + sum) and the rows the collector would produce, plus a **conclusion** pointing to the likely cause.

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Diagnostics conclusion	Cause	Fix
`No visible subscriptions`	The SP lacks `Reader`, or the tier truncates subscriptions	Assign `Reader` / review tier
`No CognitiveServices account found`	The Foundry resource is a different type or in a subscription beyond the tier limit	Verify the Foundry resource type/subscription
`AI accounts do NOT expose token metrics`	Foundry models don't emit `ProcessedPromptTokens` / `GeneratedTokens` to Azure Monitor	Cost arrives via the `CostSnapshots` fallback (Cost Management, 8–24 h latency)
`Metrics exist but return 0`	Azure Monitor latency (~15 min) or usage outside the yesterday+today window	Retry in a few minutes and re-run the `/api/cron/sync` cron
`Collector produces rows`	All OK; just needs persisting	Run the `/api/cron/sync` cron (invalidates the cache automatically)

> ℹ️ **Fix 2026-08-05:** the collector now requests each token metric **separately** from Azure Monitor. It previously requested them in a single batch and, if one didn't exist for the resource (e.g. `ProcessedInferenceTokens` on classic Azure OpenAI), Azure rejected the **whole** batch with a 400 and the panel stayed at zero.

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2. Roles and Permissions

The platform separates **two independent concepts** that combine but don't replace each other:

2.1. Role (what you can DO)

Role	Capability
SuperAdmin	Full administration of CSCloudSolutions: create tenants, configure payment gateways. Internal role, not for customers.
Owner	Tenant owner. Full access, including plan changes and billing.
Admin	Complete financial visibility, configuration changes, corrective actions (shutting down VMs, deleting resources).
Colaborador (Collaborator)	Sees financial intelligence and visibility; can suggest changes but doesn't manage billing or users.
Reader	Read-only in dashboards and reports. Cannot apply changes or view sensitive configuration.

2.2. Domain permissions (which pages you SEE)

Beyond the role, each user can be assigned one or more **domain permissions**, which determine which sidebar sections they see — independent of their role:

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Permission	Who benefits	What it enables to view
FinOps	FinOps Analyst	Cost and recommendation pages
CloudAdmin	Cloud Administrator	Infrastructure execution/write pages
Security	Security Auditor	Compliance auditing, credentials, governance
ProductOwner	Project/product lead	Visibility by Cost Center or application

How they combine: a user with the **Reader** role and the **FinOps** permission can *view* FinOps pages, but can't modify or delete anything in them — the role still governs actions. Permission filtering is opt-in: if a user has no permission assigned, they see everything their role and tier allow (default behavior, no domain restriction). **Admin/Owner** roles always see everything their tier allows, regardless of assigned permissions. The **Dashboard** (`/`) and **Support** (`/support`) are always visible to everyone.

How to assign permissions: go to `/admin/users` → select the user → toggle the permission switches in the table, or assign them when manually creating a new user. (Note: users bulk-imported from Entra ID have no permissions assigned by default — assign them afterward from the main table.)

> ⚠️ Domain permissions **do not** enable write actions on the backend — that remains exclusively controlled by the role. A user with the `CloudAdmin` permission but `Reader` role can *view* those pages, but any write action will return a 403 error if their role doesn't authorize it.

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3. Main Navigation

The left sidebar groups everything into 5 pillars (collapsible sections — the active page's section auto-expands):

- **Visibility** — dashboards, academy, FinOps maturity
- **Financial Intelligence** — billing, budgets, optimization
- **Cloud Cleanup** — zombie resources, TTL
- **Governance** — tags, policies, approvals
- **Administration** — users, configuration, billing, API

A **search bar** at the top of the sidebar lets you find pages by name or content without manually navigating categories.

3.1. "History" button — time evolution of any metric

On Dashboard, Commitment Discounts, Rightsizing, Anomalies, Budgets, and High Availability you'll find a **History** button in the top right corner. When opened:

1. Choose a date range (up to **1 year back**).
2. See the daily evolution of that page's metrics as a **line chart** and **table**.

The platform automatically saves a daily snapshot of every page (retention ~13 months) — you don't need to enable anything, it's already running in the background.

3.2. Your profile

Click your **avatar** (circle with your initial, top right) to open:

- **Full name** — editable via the pencil icon.
- **Email and role** within the tenant (read-only).
- **Display currency** — currency selector.

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- **Appearance** — Light / Dark / Automatic (follows the OS theme).
- **Sign out.**

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4. Visibility Section

4.1. Dashboard (White Board)

Your executive landing panel. Summarizes Total Potential Savings, detected Zombie Resources, and Governance rating at a glance.

How to use it:

1. Select the period at the top (current month, last 3/12 months).
2. Filter by subscription if you have several connected.
3. Pin/unpin cards for what you want to always keep at hand (pin icon on each card).
4. Any card locked by your current tier appears blurred — click it to see an upgrade modal with the detail of what it unlocks.

4.2. Costs and Projection (`/intelligence/cost-projection`, Professional+ tier)

Combines two tools in one page:

- **Cost Histogram:** daily spending distribution, with a range selector from the last month up to **13 months back** (everything Azure Cost Management allows querying), cached in Redis for instant response.
- **Cost Projection:** calculate how much you'll spend in the future.
    1. The system takes the monthly average of the last 12 months as a base.
    2. You enter an **expected annual growth %** (negative values allowed to simulate an optimization/savings scenario).
    3. Choose the horizon: 3, 6, 12, or 24 months.
    4. The result shows: base average, equivalent monthly rate, and projected total, with a real vs. projected line chart.

There's a summary card of this on the Dashboard with a **"See full detail"** link.

4.3. Azure Advisor

Direct sync with Microsoft's native recommendations, classified into Cost, Security, and Operational Excellence. Recommendations display in the language you have active on the platform (not Azure's original language).

How to use it: filter by category, review each recommendation's potential savings impact, and apply it directly from the platform or dismiss it with a justification if it doesn't apply to your case.

4.4. FinOps Maturity

Interactive assessment that places your organization at Crawl / Walk / Run across 5 pillars (visibility, optimization, governance, intelligence, operations). Answer the questionnaire to get a score per pillar plus a personalized improvement roadmap.

4.5. FinOps Academy

Interactive learning center on FinOps and Azure cost optimization — structured courses, videos, glossary, and downloadable resources.

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> ⚠️ **Mandatory gate:** new users must complete the Academy before accessing the rest of the platform (with a disclaimer explaining why). Progress is tracked **per user**, not per organization — every new person on the tenant has to complete it, and once completed it can't be re-run. Visible to all roles except SuperAdmin.

4.6-4.10. Historical Progress, TOP Expenses, Resources, Green FinOps, Captured Savings, Financial Leaks (Professional/Business+)

- **Historical Progress:** monthly trend charts, savings milestones reached, month-over-month and year-over-year comparisons. You can add manual annotations explaining a change (e.g., "migrated to reservations").
- **TOP Expenses:** ranking of your largest spending sources by service, subscription, or resource group, with drilldown to individual resource level.
- **Resources** (Business+): complete inventory — advanced filtering and search, bulk tagging, export.
- **Green FinOps** (Professional+): carbon footprint estimate by service, comparison of most efficient regions, sustainability recommendations.
- **Captured Savings** (Professional+): tracking of real savings already achieved via RIs, Savings Plans, and Hybrid Benefit, with ROI calculation.
- **Financial Leaks** (Professional+): identifies wasted money (abandoned resources, over-sizing, unnecessary redundancy) with a remediation calendar.

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5. Financial Intelligence Section

5.1. Real Consumption (`/intelligence/billing` , Professional+)

The real-time, detailed billing dashboard, straight from Azure.

How to use it:

1. Filter by period, subscription, and resource group.
2. Zoom into a specific service to see its breakdown.
3. Download the invoice or export data to Excel from the corresponding button.
4. Create threshold alerts from the same page (takes you to the Self-Service Alerts form with context pre-filled).

5.2. Budgets (`/intelligence/budgets` , Professional+)

1. **Create a budget:** name, period (monthly/quarterly/annual), limit in $.
2. **Alert threshold:** define at what % of the budget you want to be notified (e.g., 75%).
3. The system tracks automatically — see real consumption vs. budget in real time, with history of previous periods.

5.3. Cost Groups (`/intelligence/cost-groups` , Business+)

Custom cost grouping according to your own business logic (by project, line of business, application, or environment).

1. Create a group with naming rules (what resources belong, by name pattern or tag).
2. Manually assign resources if adjustment is needed.
3. Clicking a group opens a modal with tabs: **Costs**, **Actions**, **Resources**, **Governance** — each with detail specific to that group.
4. Compare groups against each other from the list view.

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5.4. Active Reservations — Commitment Discounts (`/intelligence/commitments` , Enterprise+)

Besides overall coverage and utilization, the **Active Reservations** table replicates Azure's *Reservations* blade: Name, Status, Expiration, Scope, Type, Product, Region, Renewal, Quantity, and last-day and last-7-day utilization.

- Click the **Renewal** button → modal to **enable/disable auto-renewal** for that reservation. The change applies directly in Azure — requires tenant Admin/Owner role **and** `Reservations Contributor/Owner` permissions in Azure.
- Click any **utilization percentage** → modal with last-day / 7-day / 30-day detail and daily trend.

5.5. Savings Plan vs Reservation (`/intelligence/commitment-simulator` , Professional+)

Simulate both purchase options with your real numbers and compare 1-year and 3-year savings before committing.

5.6. What-If Simulator (`/intelligence/simulator` , Enterprise tier for save/compare)

Simulate the impact of scaling compute/storage, varying network traffic, or enabling Azure Hybrid Benefit on your current cost — **before applying the real change**.

Full flow:

1. Move the sliders: compute scale, storage scale, network traffic increase %, enable/disable AHB.
2. Click **"Run Simulation"** — see the breakdown: base cost, projected cost, delta, and % change.
3. **Save the scenario:** click **"Save current"** → give it a name and optional notes. The scenario is frozen with those numbers (it won't recalculate later even if your real base cost changes over time — this lets you compare saved scenarios from different moments consistently).
4. Repeat 2-3 times with different sliders to have several saved scenarios.
5. **Compare:** check 2 to 4 scenarios with the checkboxes → click **"Compare"** → a side-by-side modal opens with each scenario as a card, marking which is the baseline.
6. **Export:** choose the format (CSV, PDF, or Markdown) next to the download buttons — you can export a single scenario, all saved ones, or the full comparison with each one's delta against the baseline.
7. **Delete:** trash icon on each row (only the creator or an Admin/Owner can delete it).

> The assumed cost mix is Azure-first: 60% of cost is compute, 25% storage, 15% network; AHB applies a flat 18% discount on the total if enabled.

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5.7-5.29. Other Financial Intelligence modules

Module	Tier	What it's for and how to use it
Cost Center Budget	Enterprise	Define cost centers, assign resources (manual or bulk), and the system automatically allocates spending to generate internal departmental chargeback.
Network Analysis	Business	Bandwidth, gateway, load balancer, and public IP costs; identifies transfer spikes and idle IPs to decommission.
Hybrid Benefit (AHB)	Business	Shows which VMs could use Software Assurance licenses and how much you'd save by enabling it; tracks what's already using it.
AKS Control	Enterprise	Active nodes, real utilization vs. over-provisioning, cost per pod, auto-scaling recommendations.
AKS Chargeback	Enterprise	Assign namespaces to teams and the system calculates how much each team spends on the cluster, for internal billing.
Container Apps	Business	Azure Container Apps cost control: monthly cost per app, environment, CPU/memory and replicas. Detects *scale-to-zero* opportunities (apps with a minimum replica ≥ 1 that could shut down when idle) and estimates the potential savings. Also available as a White Board card.
Log Analytics	Business	Log Analytics Workspaces cost control: monthly cost, retention and estimated ingestion per workspace. Detects unnecessary bulk ingestion (workspaces without a daily cap), excessive retention and *Commitment Tier* opportunities, with estimated potential savings. Also available as a White Board card.
Unit Economics	Enterprise	Define your own unit metric (cost per transaction, per user, per MB processed) and the system automatically calculates unit cost from your Azure data.
Users and Licenses (merged with "Licenses")	Professional	3 tabs: **Dashboard** (M365/Entra ID KPIs), **User Activity** (filterable table), **License Optimization** (resources without Hybrid Benefit via Resource Graph + per-SKU metrics via Microsoft Graph).
CSV Ingestion	Business	Upload a CSV with third-party billing under the FOCUS standard to analyze alongside your Azure data.
Self-Service Alerts	Business/Professional	Create budget or anomaly alert rules yourself, without asking support — condition + notification channel.
Scorecard	Business	0-100 score of your financial health with individual indicators (savings, reserve coverage, efficiency); you can set targets.
Tenant Health	Business	Overall status — detected issues, billing consistency, reservation coverage, policy compliance.
Rightsizing (+ VMSS/App Service/SQL/Storage verticals)	Enterprise	Analyzes 30 days of real usage and recommends the optimal SKU; shows estimated savings before approving the change. Dedicated verticals at `/intelligence/rightsizing/{vmss,appservice,sqldb,storage}`.
Storage Efficiency	Business	Simulates the savings from moving blobs between Hot/Cool/Archive tiers before applying it.
Compute $/Core	Professional	Cost-per-vCPU breakdown to compare VM families against each other.
Anomaly Detection	Enterprise	Automatically identifies abnormal spending spikes or drops and lets you create alerts by adjusting sensitivity.

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Module	Tier	What it's for and how to use it
Optimization Index (COIN)	Enterprise	Composite 0-100 score, comparable against industry benchmarks.
Compute Efficiency	Enterprise	Real CPU/memory/disk utilization per VM, with resize recommendation and calculated savings.
Rate Optimization	Enterprise	Compares your current rates against market benchmarks to prepare a negotiation with Microsoft.
Zero Cost	Everyone	Which resources are free in your subscription (free tier, credits) to maximize their use.
Allocation	Enterprise	Rules for distributing shared costs across multiple areas (by real usage, proportional, or fixed).
MACC Tracking	Enterprise	Tracking of the minimum annual commitment (EA/MCA) — consumed vs. committed, with compliance projection.
AI Cost Analytics	Enterprise	Cost per AI model and token consumption in Microsoft Foundry / Azure OpenAI, with call optimization recommendations.

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6. Cloud Cleanup Section

6.1. Zombie Resources (`/cleanup/zombies` , Professional+; remediation Business+)

Detects orphaned resources generating unnecessary spend: unattached disks, unused public IPs, empty App Service Plans, VMs disconnected for 30+ days.

Usage flow:

1. List with filter by resource type.
2. Review each resource's last recorded use.
3. Optional: take a snapshot of the resource before touching anything (in case you need to recover it later).
4. **Delete** — requires Business+ role and Azure deletion permissions (see the script's role table in section 1.3).
5. You can create an **auto-cleanup policy** so zombie resources of a certain type get flagged or deleted automatically going forward.

6.2. Networking Zombies (`/cleanup/zombies/networking` , Professional+; remediation Business+)

Same as above but focused on network resources: empty Load Balancers, unassociated NSGs, orphaned Public IPs, VPN gateways with no active connections.

6.3. TTL Expirations (Business+)

Control of ephemeral environments (sandboxes, test environments) with an expiration date.

1. Create a TTL policy: what resource type, how many days of life.
2. Tag affected resources with the expiration date (manually or automatically by rule).
3. The system alerts you before automatic deletion.
4. Check the history of what was deleted and when.

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7. Governance Section

7.1. Tag Compliance (`/governance/tags` , Professional+; remediation Business+)

1. Define your organization's mandatory tags (e.g., `CostCenter` , `Owner` , `Environment`).
2. The system audits your entire infrastructure and shows you which resources lack them.
3. With **auto-tagging** (Business+) you can automatically apply missing tags based on rules.
4. Generate compliance reports to show internal audit.

7.2. Governance Reporting (`/governance/reporting` , Enterprise+)

Unified executive dashboard: overall compliance status, security and tagging findings, prioritized recommendations, and historical evolution. (This page merges what used to be a separate "Governance Status" page as an additional section within the same report.)

7.3. Power Schedules (`/governance/power` , Business+)

Automated VM power on/off routines outside productive hours.

Two modes:

- **Single date:** executes the action (power on/off/restart) **once** at the exact date and time.
- **Recurring (range):** define a **"From–To"** time range and the days of the week (e.g., Mon-Fri 08:00-20:00). The system automatically creates a power-on schedule at the "From" time and a power-off schedule at the "To" time, with the same days.

Important operational details:

- The timezone is auto-detected from your browser when you open the form — you can change it manually if you need a different one.
- The system checks pending schedules every **2 minutes**, plus an immediate check when you save. The VM action can take an additional 20-40 seconds to confirm against Azure.

7.4. High Availability (`/governance/ha` , Business+)

Detects production VMs without an assigned Availability Zone or Availability Set — single point of failure risk.

7.5. Expiring Credentials (`/governance/credentials` , Business+)

Proactive alert for App Registrations / Service Principals whose secrets or certificates expire in 30/60/90 days. Each credential shows status: **Expired**, **Expiring soon** (≤30 days), or **Enabled**.

How to create an alert:

1. **"Create expiration alert"** button.
2. Define how many days in advance you want the notice (1-365).
3. Choose the channel: email, Slack, or Teams (via webhook).
4. The system evaluates daily and sends **at most one notification per day** while there are credentials within the threshold (including already-expired ones).

These rules can also be managed from **Self-Service Alerts**, under the "Credential expiration" type.

7.6. Auto-Block Policies (Enterprise+)

Automatic policies that block actions before they happen: creating VMs above a certain size, creating resources without a mandatory tag, exceeding a maximum daily spend per subscription, or creating resources outside an allowed schedule.

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7.7. Approvals (Business+)

Approval flow for infrastructure changes: a user requests the change, a specialist reviews and approves/rejects with comments, and everything is audited (who approved what and when).

---

8. Administration Section

8.1. Support (`/support` , all plans from Professional)

Any tenant user can open tickets to CSCloudSolutions and follow the conversation within the platform.

How to create a ticket:

1. Fill in subject, category (Technical / Billing / Inquiry / Feature Request), priority, and initial message.
2. You can attach screenshots or files (`jpg` , `jpeg` , `png` , `txt` , `json` ; max 5 MB per file, 10 per ticket) — kept for 60 days and then auto-deleted.
3. Support team replies appear marked with 🛟 in the thread. You can reply while it's open, and close/reopen it yourself.
4. **Quick access:** the lifebuoy icon next to your username in the header opens Support from any page. When your ticket gets a reply, you'll see a notification in the bell 🔔.

Quotas by plan:

Plan	Tickets/month	First-response SLA
Professional	20	24 h
Business	Unlimited	8 h
Enterprise	Unlimited	4 h

Global support queue (`/superadmin/support`) — CSCloudSolutions team only

Every ticket from every tenant in one queue, with four indicators: total, **unassigned**, **SLA at risk**, and mean resolution time.

SLA at risk counts only tickets **without a first response** that have less than an hour left. A ticket that was already answered, or one waiting on the customer, does not show up here even if the clock keeps running: the ball is not with the team. The applied SLA is the one for the **owning tenant's tier**, not a single value — Professional is 24 h and Enterprise is 4 h, and the queue mixes them.

Mean resolution time averages only tickets already resolved. Including open ones would produce a number that drops when new work arrives, which is the opposite of what it measures.

Filters: status pills with global counters (not scoped to the active filter), organization, priority, and assignment (*My tickets* / *Unassigned*). The table carries number, tenant, subject, category, priority, assignee, status, SLA remaining and creation date, with resizable columns and 15/30/45/60 paging.

Per-row actions: *Attend ticket* opens the conversation; *Take* assigns it to you (and moves it to "In progress" if it was open); *Resolve* closes it as resolved. Taking a ticket **always assigns to the caller** — the client cannot assign another agent by email. **Priority** can also be changed from the queue (row selector) or from the conversation panel, without reopening the ticket.

Private internal notes. In the conversation panel, the *Private internal note* toggle leaves context for the team (diagnosis, steps taken, who to escalate to) that **the customer never sees**. The filter is server-side, not on screen: the note does not leave the backend toward a tenant user. An internal note also does not move the ticket status or count as a first response — the customer saw nothing, and the SLA reflects that.

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8.2. Users and Permissions (`/admin/users`)

See section 2 for details on role vs. permissions. From here you add users, edit their role, toggle their domain permissions, or fully deactivate them.

8.3. Configuration (`/admin/config`)

General tenant profile administration: name, logo, default language for new users, timezone for reports, billing cycle.

8.4. Billing — Plan Change (`/admin/billing`, Professional+, Owner role)

Plan Limits:

Plan	Allowed Azure Subscriptions	Users per Tenant	Support / SLA
Professional	Up to 2 subscriptions	Up to 3 users	20 tickets/mo (24 h)
Business	Up to 3 subscriptions	Up to 5 users	Priority (12 h)
Enterprise	Unlimited	Unlimited	Dedicated 24/7 (99.9% SLA)

Change procedure:

1. Choose the new plan (Professional / Business / Enterprise).
2. Choose frequency (monthly/annual) and proration mode.
3. The system shows a **preview summary** with the real amount calculated by the payment gateway before confirming: *"You'll be charged $X now"* (upgrade) or *"You'll receive a $X credit"* (downgrade), the new recurring total, and the next billing date.
4. The change **only applies** when you click **Confirm change** — until then you can cancel at no cost.

8.5. Notification Configuration (`/admin/notifications`, Professional+)

Three supported channels: **Slack**, **Microsoft Teams**, **Email (SMTP)**.

Configure Slack:

1. Go to `https://api.slack.com/apps` → create or select an app → **Incoming Webhooks**.
2. **"Add New Webhook to Workspace"** → choose the destination channel.
3. Copy the webhook URL (starts with `https://hooks.slack.com/services/...`).
4. In the platform: **Notifications** → **Add Channel** → **Slack** → paste the URL → save.

Configure Microsoft Teams:

1. In the Teams channel → ⋯ → **Connectors** → search for "Incoming Webhook".
2. Configure the webhook and give it a name.
3. Copy the generated URL.
4. In the platform: **Notifications** → **Add Channel** → **Microsoft Teams** → paste the URL → save.

Configure Email:

- Uses SMTP. If your organization doesn't specify its own server, the platform's default SMTP configuration is used.
- You can define multiple recipients and, optionally, override host/port/user/password per channel.

Severity filter: each channel can filter what it receives — `info`, `warning`, `error`, or combinations (e.g., `warning,error` to skip informational messages).

Test the channel: a test button on each configured channel — sends a test notification before you rely on it in production.

Tier availability: Professional allows 2+ channels; Enterprise allows unlimited channels + per-channel SMTP overrides.

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If notifications aren't arriving, check in this order: channel enabled (`enabled = true`) → severity filter matches the alert level → you tested the channel from the UI → there's at least one channel configured for the tenant. The send log (with errors) is available for diagnosis.

8.6. Executive Report (Business+)

Automated generation of high-level periodic reports, designed for presenting to leadership — downloadable executive summary.

8.7. SaaS Operations (SuperAdmin) (`/superadmin/ops`)

Global operations control center for SaaS monitoring:

1. Check overall platform status and unacknowledged alerts.
2. Review component health and cron execution status (last run, age, summary, state).
3. Verify notification-channel coverage across SuperAdmin tenants.
4. Dispatch operational notifications to configured channels when degradation is detected.

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8.8-8.18. Other Administration modules

Module	Tier	Usage
Customer Onboarding	Admin	See section 1.3 — tenant onboarding and Service Principal credentials.
Tenant Management (Commercial)	SuperAdmin	`/admin/tenants` : manual provisioning, tier/status changes, seller/referrer tag, and commission (%) per tenant.
SaaS Operations	SuperAdmin	`/superadmin/ops` : component health + cron status + operational notification dispatch.
Azure Lighthouse Onboarding	Enterprise	Generation of ARM template for cross-tenant delegation, at `/admin/onboarding/lighthouse` .
AI Configuration	Professional	Enable/disable AI features, choose model, adjust detection sensitivity and what data is shared.
Invoicing Report	Enterprise	JSON/CSV/PBIT stub export with detail by billing profile, invoice section, and customer, at `/admin/report` .
Workbooks	Enterprise	Custom reports: own layout, charts, data tables, exportable to PDF and shareable via link.
Audit Trail	Professional	Complete log of who changed what and when, filterable by user/action/date, exportable for compliance.
MCP API Keys	Enterprise	API keys for MCP-style integrations — generate, rotate, and revoke.
Public API	Enterprise	Documented REST API (OpenAPI/Swagger), with rate limits and usage monitoring.
Power BI Templates	Enterprise	Downloadable `.pbit` templates, pre-connected to your SaaS data.
FOCUS 1.1 Export	Professional	Export your data in the standard FOCUS format, schedulable for automatic daily generation.
SSO SAML	Enterprise	Configure your Identity Provider, map attributes (roles, emails), test, and activate for the whole organization.
Partner Markup (CSP)	Enterprise	Configurable margins per service for CSP partners, automatically applied in billing.
M365 Copilot	Enterprise	Tenant configuration + assisted chat over your own FinOps data, at `/admin/copilot-m365` .

9. FinOps Copilot (AI Assistant)

Floating icon in the bottom corner of the screen, available on Professional+.

- **Automatic context awareness:** the Copilot reads the content of the page you're on — you don't need to tell it which module you're in. When you open it, without typing anything, it generates an **executive report** of what's being shown: module context, key findings, savings opportunities prioritized by impact, risks, and a 7-day action plan.
- **Targeted questions:** besides the automatic report, you can ask it directly. E.g., in Budgets: *"Summarize the current state of our budgets"*.
- **Corrective actions:** with your prior authorization, it can guide you through deleting zombie resources or applying missing tags via automated scripts — it never executes anything without your confirmation.

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10. Account Security (MFA)

TOTP-based two-factor authentication (Google Authenticator, Microsoft Authenticator, Authy, etc.), optional per user, required only for **sensitive operations** (deleting a tenant, canceling a subscription, changing billing configuration).

How to enable it:

1. Go to your profile → **Security Settings** → **"Enable 2FA"**.
2. The system shows you a QR code — scan it with your authenticator app.
3. Enter the 6-digit code shown by the app to confirm.
4. You receive **10 recovery codes** — download and store them somewhere safe. They're shown only once.

When it's requested: when you attempt an operation marked as sensitive, a modal appears asking for your 6-digit code (or a recovery code if you've lost access to the authenticator). Each recovery code is single-use; if you use up all 10, you must disable and re-enable 2FA to generate a new set.

If you lose access to your authenticator: use any of your 10 recovery codes to log in and reconfigure 2FA from scratch. If you've also lost the recovery codes, an Admin can force-disable your MFA (logged in the audit trail).

---

11. Advanced Features and Integrations

This section is for technical users (Cloud Admin, DevOps) who need to integrate the platform with other tools or configure advanced access.

11.1. SSO SAML (`/admin/sso`, Enterprise, via WorkOS)

Lets Enterprise customer users log in with their own Identity Provider (Okta, Auth0, AD FS) instead of — or alongside — Microsoft Entra ID.

Configuration per customer:

1. Go to **Admin → SSO SAML** (only visible on Enterprise tier).
2. Get the `workos_org_id` and `workos_connection_id` from the WorkOS dashboard (CSCloudSolutions creates the Organization/Connection there if they don't exist).
3. Fill in the form: **Domain** (e.g., `acme.com`), **WorkOS Organization ID**, **WorkOS Connection ID** → enable the **Enable SSO** toggle → save.
4. **"Generate Admin Portal"** button → opens a WorkOS link in a new tab; the customer's IT admin receives an email to configure their own IdP.
5. (Optional) **"Test SSO"** button → redirects you to the IdP to validate the login works before announcing it to end users.

How the end user logs in: on the login screen, they choose "Login with SSO" instead of "Sign in with Microsoft" → the system redirects them to their own IdP → after authenticating, they're automatically returned to the dashboard.

> ⚠️ Only **one IdP connection per tenant** is supported by current design. SSO coexists with Microsoft Entra ID login (MSAL) without replacing it — both methods work simultaneously, so you don't lose existing access when enabling SSO. The SSO session lasts 12 hours.

Common errors: "SSO not configured" (platform credentials missing), "SSO not enabled for this tenant" (the toggle is off), "Missing workos_connection_id" (the portal hasn't been generated yet, or the customer hasn't configured their IdP).

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11.2. Audit Trail — advanced usage (`/admin/audit`, Professional+)

Beyond the filterable log (by email, action type, status, and date range, 10 rows per page), you have 4 export formats: **CSV (current page)**, **Full filtered CSV (streaming)**, **JSON**, **NDJSON**.

Typical use cases:

- **Compliance audit (SOC2):** export the full quarterly range with the filtered export button.
- **Investigate an incident:** filter by the suspicious user's email + `FAILURE` status.
- **Operational monitoring:** filter by action type `DELETE` + a recent date range to see what was deleted.

> ⚠️ The full export has a **100,000-row cap** — for larger datasets, narrow the date range and export in parts. Log retention is currently indefinite (no automatic purge yet).

11.3. Public REST API v1 (`/admin/api-keys`, Enterprise)

Read-only programmatic access to your data (costs, budgets, recommendations, anomalies) to connect BI tools or your own scripts.

How to generate and use it:

1. **Admin → Public API** → generate a new key, choosing only the **scopes** you need (`read:cost`, `read:resources`, `read:budgets`, `read:recommendations`, `read:anomalies`) — request the minimum necessary, e.g., `read:cost` if it's only to feed a BI dashboard.
2. Authenticate each request with the header `X-API-Key: pak_live_xxx` (or `Authorization: Bearer pak_live_xxx`).
3. Example endpoints: `GET /cost/summary?from=...&to=...&groupBy=service`, `GET /cost/timeseries?granularity=daily|monthly`, `GET /resources?type=...&limit=100`, `GET /budgets`, `GET /recommendations`, `GET /anomalies?severity=low|medium|high`.
4. Interactive documentation (Swagger) available at `/api/v1/docs`.

> ⚠️ Default limit: **60 requests/minute** per key (`X-RateLimit-*` headers show your remaining quota; hitting the limit returns a 429 error). If you need more for a batch job, request it from Admin → Public API or via support (up to 1000 req/min). Monetary values always travel as **strings** (e.g., `"1234.56"`), not numbers — parse them as text to avoid losing precision.

11.4. Showback/Chargeback PDF Export (`/admin/report`, Business+)

Generate and email showback/chargeback invoices to each customer or internal cost center.

- **Download a single invoice:** from the billing report, select the customer/center and period → the PDF downloads directly.
- **Download all at once:** same flow without selecting a specific customer → a ZIP with all the period's invoices downloads.
- **Send via email:** fill in the recipient's email and name → it's automatically sent with the PDF attached via Microsoft 365 (requires CSCloudSolutions to have email sending configured for your tenant — if unavailable you'll see a clear error asking you to contact support).

Every send and download is logged in the audit trail.

11.5. FOCUS 1.1 Export — advanced usage

Beyond manual download from `/admin/focus-export` (see section 8.7), you can automate extraction with your own MCP API Key (generated at `/admin/mcp-keys`):

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```
curl -H "Authorization: Bearer mcp_xxx" \
  ".../api/exports/focus?tenantId=...&format=ndjson"
```

Available formats: CSV, JSON, NDJSON. Cap of **500,000 rows** per export (default 100,000). Useful for automatically feeding external FinOps tools (Power BI, CloudHealth, etc.) without manual intervention.

11.6. Status Page

A public page (no login required) where you and your users can check at any time whether the platform is operational: `/es/status` (or `/en/status`, `/status`). Shows the status of API, database, Azure sync, AI provider, and billing, plus an incident history. Useful to share with your team if something seems broken — before opening a ticket, check whether an incident is already reported there.

11.7. Paddle Billing — technical detail

The plan change (see section 8.4) has 3 possible proration modes when upgrading/downgrading:

- **Immediate proration:** the difference is charged or credited right now.
- **Next-cycle proration:** the full price change is charged only at the next renewal.
- **No immediate charge:** the plan change applies without generating any charge until the next cycle.

From `/admin/billing` you also access the **payment management portal** (update card) and the **downloadable invoice history**.

11.8. 7-day free trial (self-service)

If you signed up yourself from the pricing page (not through a SuperAdmin's onboarding), your account starts with a **7-day trial** on the plan you chose. You'll see a trial status banner at the top of the platform:

- 🔵 Blue: 5 or more days remaining.
- 🟡 Yellow: between 3 and 4 days remaining.
- 🔴 Red: 2 days or fewer — this one can't be dismissed.

You can upgrade to a paid plan at any time from **Billing** — the trial immediately converts to an active subscription. If the trial expires without an upgrade, the account switches to limited access until you activate a paid plan.

---

12. Best Practices

- **Weekly review:** check the **Dashboard** and **Zombie Resources** at least once a week to catch financial leaks before they pile up.
- **Automate early:** enable **Power Schedules** on your Development/Testing environments as a first step — it's common to see 60% savings in idle compute hours with no other change.
- **Enforce tag compliance:** without consistent tags, the chargeback/showback module can't fairly distribute the monthly bill across teams — it's the foundation everything else relies on.
- **Use the What-If Simulator before committing:** before purchasing a Reservation or Savings Plan, simulate the scenario and save it — it gives you a concrete number to justify the decision to finance.
- **Set up at least one notification channel from day one** (Slack/Teams if your team already lives there, or email if you prefer simplicity) — budget alerts are useless if nobody sees them in time.

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Support and Contact

For any additional assistance, open a ticket from **Support** (`/support`) within the platform, or write to **soporte@cscloudsolutions.com.ar**.

---

13. Global Communications

Where: sidebar → **Global Communications** (`/superadmin/announcements`). SuperAdmin only.

Use it to tell tenants something from inside the platform: scheduled maintenance, an incident, a change of terms.

13.1 Creating an announcement

Press **New announcement** and fill in:

Field	Notes
Title and message	The message supports Markdown (bold, lists, links)
Severity	Info / Maintenance / Warning / Critical — sets the banner color
Channels	**Banner** (top strip), **Popup** (window), or both
Validity	From / to. The notice appears and disappears on its own within that window
Scope	All tenants, or a selection
Action link	Optional. Adds a "More information" link

Save draft leaves it ready without publishing; **Publish** puts it into effect according to its dates.

13.2 Specific scope

Unchecking *"All tenants"* opens a searchable tenant list — filter by name or ID, each showing its status (ACTIVE, TRIAL, PAST_DUE).

CANCELED tenants don't appear: their users no longer sign in, so targeting them has no effect. TRIAL and PAST_DUE do appear — a PAST_DUE is often exactly who you want to notify.

If you edit an old announcement targeting a tenant no longer on the list, it shows as **"Not in list"**. That's deliberate: you can uncheck it instead of silently losing it on save.

13.3 Translations (optional)

The form has a collapsible **Translations** section with fields for English and Portuguese.

They are not mandatory. If you only fill in Spanish, users in other languages see the notice in Spanish. That's deliberate: an outage notice shouldn't wait for a translation.

A half-finished translation (title only, or message only) is **discarded on save** — showing a blank body would be worse than showing the full text in another language.

The **Languages** column shows at a glance which ones each announcement has.

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13.4 Statuses

Status is computed from the dates:

Status	Meaning
Draft	Saved, not published
Scheduled	Published, start date not reached yet
Active	Showing right now
Finished	Past its end date
Cancelled	Manually withdrawn

Cancel (⏹) pulls it out of circulation immediately without deleting it. **Delete** (🗑) removes it along with its dismissal records; it cannot be undone.

---

14. Tenant Lifecycle

The **Tenants** view has three new columns, hidden by default — enable them from the column picker:

- **Activation Date** — when it was effectively activated
- **Cancellation Date** — when it was cancelled, if applicable
- **Cancellation Reason** — Voluntary churn / Payment delinquency / Contract expired / Admin deprovisioning

These dates are now recorded automatically: when a cancellation arrives from the payment gateway or the marketplace, when a contract expires, or when an account is suspended for non-payment. A **full history** of every transition is also kept, so a tenant that cancelled and came back retains all its periods — which is what cohort retention analysis needs.

> **About tenants created before September 2026:** their "Activation Date" is the record creation date, not the effective onboarding or first payment. That distinction wasn't recorded at the time and cannot be recovered. Treat them as approximate.

14.1 Filtering and export

Above the table you can pick **Activations** or **Cancellations** and set a date range. The filter stacks with the search box, so you can combine both.

Export CSV downloads exactly what you're seeing — filters applied — including a **tenure in months** column (activation → cancellation, or activation → today if still active). It's built for the accounting report: opens correctly in Excel, with proper accents.

15. Azure Marketplace

The platform can also be sold through the Azure Marketplace. The technical flow is already built; what's left to publish is in `docs/marketplace-publicacion-checklist.md`.

15.1 What happens on purchase

1. Azure sends the buyer to the landing page with a temporary token.

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2. The platform resolves that token against Microsoft and shows the purchased plan.
3. The customer signs in with Entra ID and activates.
4. Microsoft starts billing and notifies us by webhook of every change: plan change, quantity, suspension, reinstatement and cancellation.

15.2 Two statuses that are NOT the same

The tenant record shows two similar fields, and the difference matters:

Field	What it is
Marketplace status	What **Microsoft** says: Subscribed, Suspended, Unsubscribed
Subscription status	**Our** commercial status, also driven by Paddle and your actions in this panel

They're kept apart deliberately. As a single field, a suspension originating in Azure would erase the real reason for a cancellation handled through another channel — and in a billing dispute you couldn't explain where each status came from.

15.3 Reconciliation data

For each Marketplace subscription we store the offer, the purchaser's email and the directory they purchased from. **The purchaser isn't always the person who later uses the platform**, and that distinction is what Microsoft asks for to resolve a dispute.

---

16. Onboarding via Azure Lighthouse (Enterprise)

This is the alternative to the onboarding script: the customer **delegates** access to their subscriptions from their own directory, without handing you credentials or running anything in their environment. It lives under **Users & Access → Lighthouse Onboarding**, and only appears on the **Enterprise** plan.

16.1 When it fits and when it doesn't

Good fit	Poor fit
The customer won't run scripts in their environment	You need to read secrets from their Key Vault or blobs from their storage
The customer wants to be able to revoke access at any time	The customer manages spend by management group
You'd rather not maintain one secret per customer	The customer has a single subscription and already ran the script

Lighthouse delegates the **control plane only**. There is no access to the customer's data plane, and their management groups are not delegated — delegation is per subscription.

16.2 The procedure

1. **Fill in both fields.** *Managed Tenant ID* is the GUID of the customer's directory; *Managed Subscription ID* is the GUID of one of their subscriptions.
2. **Pick the roles.** For a visibility-only onboarding, **Reader** and **Cost Management Reader** are enough. *Contributor* grants write access over every resource in the subscription: ask for it only if you are going to operate the customer's infrastructure.

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3. **Generate the ARM template** and download or copy it.
4. **The customer deploys it** in the Azure portal, inside **their own directory**: *Deploy a custom template → Build your own template → paste the JSON*. They need to be Owner of that subscription.
5. **Verify the delegation** with the button in the panel.

16.3 Step 5 is not optional

Generating the template and having the customer deploy it are two different things. The row stays at **"Pending acceptance — Not confirmed in Azure"** until you verify, and until that moment the platform **keeps accessing through the previous model**: the delegation exists in Azure and goes unused.

Verify delegation asks Azure, and it is the only thing that changes the tenant's access mode. If it reports active, open the customer's cost dashboards to confirm data is arriving.

16.4 A customer with several subscriptions

You don't need to generate one template per subscription. The same template delegates *whichever subscription it is deployed into*, so the customer applies it once in each one, switching context between deployments.

> The *Managed Subscription ID* field **does not target the template**: it only feeds our record. If the customer deploys it into a different subscription, that other one is what gets delegated. That is what makes the template reusable, but it's worth knowing.

Delegated subscriptions show up on their own at the next sync cycle; there is no need to register them by hand.

16.5 The two rows and the Delete button

Once a delegation is active you will see **two** entries for the same subscription: one reported by Azure and one that is our own record. That is expected.

Delete removes **our record**, not the delegation. The `registrationAssignment` lives in the customer's subscription and can only be removed from their directory. If you delete one that was still active, the warning tells you so: the customer keeps delegating access that the platform no longer records.

For the rows Azure reports, the button rejects them with the reason — they are not our records.

> If you remove the **last** delegation for a tenant, it automatically reverts to the App Registration model. That is deliberate: with no delegation behind it, a tenant in Lighthouse mode would be left with no route into Azure at all.

16.6 If something doesn't work

Message	What's going on
"AZURE_LIGHTHOUSE_PRINCIPAL_ID is not configured"	Platform configuration is incomplete. Without it the template would deploy successfully and grant access to nobody: that's why it is blocked beforehand. Escalate to infrastructure.
"A subscription cannot be delegated to the directory it already belongs to"	The *Managed Tenant ID* is ours. Lighthouse exists so that the **customer's** directory delegates into ours.
"Available on the Enterprise plan only"	The tenant isn't Enterprise. On Professional and Business, onboarding is done with the script.
Delegation active but no data	Check the delegated roles: without *Cost Management Reader*, no consumption data arrives.

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17. Affiliate Program

Path: `/superadmin/affiliates` · **Menu:** Administration → Affiliate Program

An affiliate is a third party — a reseller, influencer or partner — who promotes the platform with their own link and earns a percentage of what the customers they brought in pay.

An affiliate **is not a user of any tenant** and has no account on the platform: they get their own read-only dashboard through a personal link that you generate (17.5). There they see their referred customers and their commissions, and nothing else — not the cloud spend of the customers they brought in, and nothing about other affiliates.

> **Who does what, at a glance**
>
> | You, here | The partner, on their own |
> |---|---|
> | Register them and set their % | Shares their referral link |
> | Generate their dashboard link | Opens the dashboard to see their numbers |
> | Invite them to Payoneer (17.6) | Completes their Payoneer registration |
> | Build the run and transfer (17.7) | Invoices you, where applicable |

17.1 Registering a partner

1. **New affiliate**.
2. Fill in name, email and **referral code**. The code goes in the URL, so it accepts letters, numbers, hyphens and underscores, and must start with a letter or number. It has to be unique.
3. **Commission (%)** — 20 % by default. This is the percentage applied to every charge made to that affiliate's customers.
4. *Payout method* and *Payout reference* are free text for whoever settles payments (bank transfer, PayPal, alias). No structured bank details are stored.
5. **Save**.

If the email or the code already exists, the creation is rejected with a notice: both are unique on purpose, so two affiliates cannot claim the same attribution.

17.2 The referral link

Copy on the affiliate's row puts this on the clipboard:

```
https://finops.cscloudsolutions.com.ar/signup?ref=CODE
```

When someone arrives through that link, the code is stored in their browser for **60 days**. If they sign up within that window, the customer is attributed to that affiliate.

Three rules worth knowing when answering questions:

- **First touch wins.** A customer belongs to exactly one affiliate and is never reassigned. If they came through A's link and later through B's, the customer is A's.
- **No self-referrals.** If the affiliate's email matches a user of the customer, the attribution is rejected.
- Attribution happens **when the customer signs up**. A customer that already existed is not attributed by visiting a link later on.

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17.3 Editing, suspending and deleting

The pencil button opens the same form used for creation, pre-filled. You can change name, email, code, percentage, status and payout details.

Two things worth understanding before you touch anything:

- **Changing the percentage does not rewrite history.** Every commission stores its own snapshot of the percentage in force when it accrued. The new value applies from the next charge onwards.
- **Changing the code breaks links already shared.** The visitor's cookie stores the code, not the affiliate, so anyone arriving through the old one is left unattributed. The form warns you when you are changing it.

Suspend is what to use when you stop working with someone: it stops new commissions from accruing and keeps the entire history. The affiliate stays in the list, greyed out, with their numbers intact. **Reactivate** puts them back.

Delete only works if the affiliate has **no commissions at all**. If they have even one, pending included, the platform refuses and tells you how many there are. This is not an arbitrary restriction: given how the database is built, deleting the affiliate would take their commission history with it — including what you already paid them — and that record cannot be rebuilt. That is what *Suspend* is for.

If the affiliate has referred customers but no commissions yet, they can be deleted; the confirmation notice tells you how many customers will be left unattributed, and that cannot be undone.

17.4 Commissions and settlement

The **Accrued commissions** table fills itself: every time an attributed customer pays, a commission is recorded.

Through each customer's first year, and no further. The window is 12 months from that customer's first charge: a monthly customer accrues their twelve charges, and an annual one gets the initial charge while next year's renewal falls outside. It is measured in months rather than in number of charges precisely for that reason — "up to 12 charges" would give an annual customer twelve years.

The partner dashboard shows the cut-off date per customer, so "how long do I earn on this one?" answers itself.

Status	What it means
Pending	Accrued. Just charged to the customer; it is worth waiting out the refund window before approving.
Approved	Reviewed and ready to pay.
Paid	Already settled with the affiliate.
Reversed	The customer requested a refund and the commission was voided automatically.
Cancelled	Voided manually.

To settle: tick the rows, click **Approve**, and once the transfer goes out, **Mark as paid**.

> A **Paid** commission cannot be rolled back from the interface, and a later refund will not reverse it either. If it needs correcting, deduct it from the next settlement: that is an accounting operation, not a click.

The **Unsettled** and **Paid** columns in the partners table are totals in the currency of the most recent charge. If one affiliate has customers paying in different currencies, those totals are not comparable — check the per-commission detail.

17.5 The partner dashboard

The **dashboard** button on the affiliate's row generates their personal link:

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```
https://finops.cscloudsolutions.com.ar/en/partner#k=afp_live_...
```

It is shown only once. The database keeps nothing but its hash, so there is no way to see it again: if you lose it, you issue a new one — and that invalidates the previous link. Same deal as API keys.

What the partner sees: their totals (pending and already paid), their referral link, their referred customers with the date each one stops earning, and the detail of their commissions with each status.

What they do **not** see, worth knowing so you can answer: the cloud spend of the customers they brought in, their Azure identifiers, or anything about other affiliates. Only the company name, the dates and their own commission amounts.

The **broken-link** button revokes access without touching the affiliate or their commissions. And **suspending** an affiliate also closes their dashboard: there is no need to revoke as well.

> The link **is** the credential: whoever holds it sees those numbers. Send it to the person over a direct channel, do not paste it into a group chat.

17.6 Inviting the partner to Payoneer

> This only works if Payoneer is configured. Otherwise the button answers "Payoneer is not configured", which is not an error: the program credentials simply have not been loaded yet. See `docs/payoneer-payouts.md`.

Payoneer cannot transfer to someone who is not registered in our program. The **bank** button on the affiliate's row requests that registration and returns a sign-up link, which is also shown only once and is valid for **a single registration**.

The button colour tells you where they stand:

Colour	Means
Grey	Not invited yet
Amber	Invited; registration still pending
Green	Registered and able to receive payments

What the partner does with that link, worth telling them up front because it is what generates the most questions:

1. Opens the link. Their name and email arrive pre-filled, and they can correct them.
2. Chooses whether they are paid as an **individual** or a **company**, and completes contact details and address.
3. Uploads an **identity document**. Payoneer runs identity verification (KYC); this is not optional and we cannot skip it for them.
4. Adds the **bank account** where they want the money.
5. If they already had a Payoneer account, the flow offers to link the existing one instead of creating another.

Approval is **not immediate**: Payoneer decides and it can take time. Until they are approved you cannot transfer to them — their commissions keep accruing all the same, they just wait.

> **Tell them before they start:** Payoneer charges the recipient, and the terms vary by country. For someone in Argentina that can outweigh the convenience. If they prefer a direct transfer, the *Payout method* field on their record exists for exactly that and requires no registration anywhere.

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17.7 Payout runs

Path: `/superadmin/comisiones`

A run groups everything due in **one currency** into a batch, per beneficiary, and is what ties a transfer to the commissions it covered. Affiliates and sales reps go together: they are in the same ledger.

The monthly cycle:

1. **Pick a currency and a per-person minimum** and build the run. The minimum applies to **each beneficiary's total**, not to each commission: twelve commissions of $0.30 are $3.60, which may be worth a transfer even though none of the twelve is on its own. Anyone below it stays out and their balance carries over to the next run.
2. **Download the payment order** (CSV). One row per person with the amount, email, payout method and payout details: what you need in order to transfer, without looking up each record.
3. **Transfer**, by whatever means you agreed with each of them.
4. **Confirm the payment** with the reference. Only then do that run's commissions become *Paid*, all together and under the same reference.

Cancelling an open run releases its commissions for the next one. A run already paid cannot be cancelled: the money left, and that is fixed with an adjustment.

Three things that prevent surprises:

- **One currency per run.** You cannot transfer in two currencies at once, and the platform does not invent an exchange rate.
- **Two runs do not collide.** Commissions taken by one are reserved; a run built in parallel will not take them.
- **Confirming twice does nothing.** It is safe to retry if the connection dropped.

The minimum you used is stored on the run. Changing the default next month does not rewrite why an old run left someone out.

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